Camille Billouard

PhD in AI | 3D Computer Vision & Neural Rendering

Paris, France | Personal website | GitHub | Google Scholar

Summary

AI researcher specialised in neural 3D reconstruction from satellite imagery at city scale. I turn research prototypes into fast, memory-bounded pipelines with PyTorch, custom CUDA and HPC, and I back engineering choices with theory and careful measurement.

Technical Skills

Programming: Python, C++, CUDA, SQL, Java, Bash

ML & 3D Vision: PyTorch, NeRF (nerfstudio), Gaussian Splatting (LichtFeld Studio)

Geospatial: xDEM, LiDAR HD, Pléiades Neo

Tools: Git, Linux, SLURM / HPC

Languages: French (native), English (fluent, spoken and written)

Experience

PhD Researcher — Oct 2023 – Oct 2026
CNES (French Space Agency) & LASTIG, IGN (French Mapping Agency) — *Paris, France*

- Built **SAT-NGP**: satellite NeRF training **86–200× faster** (10–20 h down to 4–14 min), elevation error within 0.19 m of the state of the art; now a reference baseline for efficient 3D Earth observation.
- Built **Snake-NeRF**: out-of-core tiling that reconstructs **52 km²** of Pléiades Neo imagery (442 tiles) **on a single commercial 16 GB GPU**, with bounded peak memory (within 1% of the analytical prediction), 5× lower host memory and near-linear runtime.
- Formalised scalability for neural rendering (linear time, bounded memory, preserved quality) and derived a **fully scalable pipeline** on real satellite data.
- Developed custom CUDA / C++ kernels integrated into PyTorch to accelerate the full neural rendering pipeline (RPC ray generation, segmented volume rendering, ray sorting).

Research Intern — Apr 2023 – Oct 2023
CNES (French Space Agency) — *Toulouse, France*

- NeRF-based 3D reconstruction from multi-date satellite images.

Research Intern — May 2022 – Jul 2022
IRIT (Toulouse Institute of Computer Science Research) — *Toulouse, France*

- Fast 3D super-resolution for small-field dental cone-beam CT volumes.

Software Engineering Intern — Apr 2021 – Aug 2021
CGI — *Bordeaux, France*

- Java / Spring Batch jobs for banking process automation (Oracle SQL, GitLab, Sonar).

Supervision

Co-supervised two research interns during the PhD: **NeRF-based semantic segmentation** (C. Herbin) and **3D reconstruction from CO3D video sequences** (K. Specogna), the latter published in RFPT 2026.

Publications & Talks

C. Billouard, D. Derksen, A. Constantin, B. Vallet. *Tile and Slide: A New Framework for Scaling NeRF from Local to Global 3D Earth Observation.* ICCV Workshops, Hawaii, 2025. arXiv:2507.01631

C. Billouard, D. Derksen, E. Sarrazin, B. Vallet. *SAT-NGP: Unleashing Neural Graphics Primitives for Fast Relightable Transient-Free 3D Reconstruction from Satellite Imagery.* IGARSS, Athens, 2024. doi

K. Specogna, **C. Billouard**, et al. *Applications des champs de radiance neuronaux au mode vidéo de CO3D.* Revue Française de Photogrammétrie et de Télédétection (RFPT), 2026.

Talks: ICCV (Hawaii, 2025), IGARSS (Athens, 2024), ESA Living Planet Symposium, ORASIS, SFPT, Meet'Up ML (Nantes).

Education

PhD in Artificial Intelligence — 2023 – 2026
Université Gustave Eiffel, CNES, LASTIG, IGN — *Paris, France*

MSc in Artificial Intelligence: Foundations and Applications (IAFA) — 2021 – 2023
Université de Toulouse — *Toulouse, France*

BSc in Computer Science, research track — 2021
Université de Lille — *Lille, France*

BSc in Mathematics & Computer Science — 2020
Université de Bordeaux — *Bordeaux, France*